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Synergistic effects of Fe–Mn modified biochar and water management on remediation of Cd and Hg co-contaminated soils

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Abstract

Cadmium (Cd) and mercury (Hg) contamination in paddy soils presents a formidable challenge to global food security and human health, primarily due to their contrasting biogeochemical behaviors under varying water management regimes. This study investigated an integrated remediation strategy combining iron–manganese oxide modified biochar (FMBC) application with two distinct water management regimes (continuous flooding and aerobic irrigation) to synergistically mitigate Cd and Hg accumulation in rice and elucidate their underlying mechanisms, including the impacts on soil microbial communities. Pot experiments revealed that continuous flooding effectively decreased soil Cd bioavailability and promoted iron–manganese oxide plaque (IMP) formation on the root surface, thereby impeding Cd uptake. Conversely, this condition inadvertently promoted the relative abundance of *hgcA*-containing methylating microorganisms, thereby elevating the risk of Hg methylation. In contrast, aerobic irrigation reduced total Hg (THg) and methylmercury (MeHg) accumulation in rice grains, yet concurrently exacerbated Cd uptake by the plants. The incorporation of FMBC effectively resolved this antagonistic dilemma through dual immobilization mechanisms. Specifically, the combination of continuous flooding and FMBC application minimized grain Cd concentration (0.05 mg kg^{-1}), while simultaneously suppressing the flooding-induced proliferation of *hgcA*-containing microbes. Conversely, integrating aerobic irrigation with FMBC achieved the lowest grain THg (0.02 mg kg^{-1}) and MeHg (6.89 μg kg^{-1}) levels, while simultaneously restricting aerobic-induced Cd uptake. Overall, the synergistic integration of FMBC with specific water management provides a highly effective and sustainable strategy for mitigating both Cd and Hg bioaccumulation in rice grains, thereby enhancing food safety in co-contaminated paddy ecosystems.

Highlights

- FMBC integrated with flooded irrigation significantly promoted rice growth and tillering.
- Different water management regimes had opposite effects on soil bioavailable Cd and Hg.
- FMBC combined with water management synergistically reduced Cd and Hg in rice.
- FMBC suppressed flooding-induced *hgcA*-containing Hg-methylating microbes.

Keywords Cd–Hg co-contamination, Fe–Mn modified biochar, Water management, Bacterial community, *Oryza sativa* L.

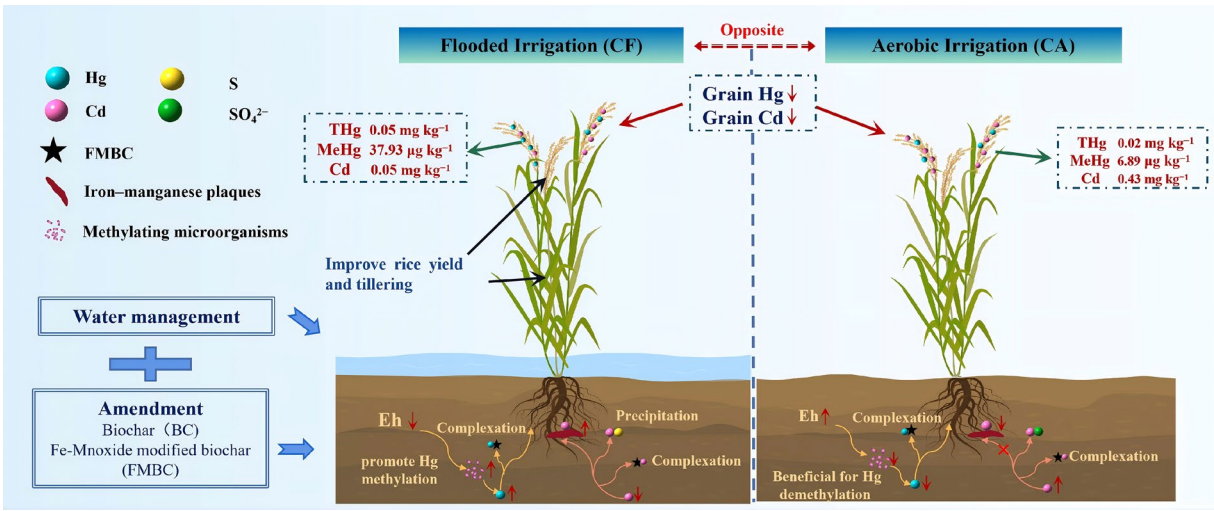
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Graphical Abstract



1 Introduction

Heavy metal contamination, particularly cadmium (Cd) and mercury (Hg), poses a pervasive and escalating global environmental threat to agricultural systems. Their persistent nature and detrimental effects on ecosystems and human health necessitate urgent and effective remediation strategies (Wan et al. 2024; Mai et al. 2025). In China, the National Soil Pollution Survey Bulletin reported that the average concentrations of Cd and Hg in soil exceeded the regulatory limits by 1.6% and 7.0%, respectively (MEP and MLR 2014). As a staple food, rice (*Oryza sativa* L.) accounts for 30% of the national grain output and 35% of global rice production (Liu et al. 2023b; Zhang et al. 2025). However, its unique flooded cultivation practices and inherent physiological traits render rice particularly vulnerable to heavy metal accumulation compared to other cereal crops (Luo et al. 2023). The uptake of Cd and Hg by rice plants significantly disrupts their physiological and metabolic processes, leading to accumulation in grains and subsequent reduction in yield and quality (Bao et al. 2025). Furthermore, these heavy metals can be transferred and biomagnified along the food chain, posing substantial public health risks through dietary exposure (Liang et al. 2025; Zhao et al. 2026). Therefore, it is necessary to implement measures to remediate paddy soils contaminated with Cd and Hg.

As a fundamental agronomic practice, water management can regulate heavy metal bioavailability by altering soil physicochemical properties, particularly the redox potential (Eh) (Li et al. 2022b). Nevertheless, owing to the differences in the biogeochemical properties of heavy metals, rice plants exhibit varying trends of Hg and Cd accumulation under different water management practices (Zhao and Wang 2020). Flooding in paddy fields induces anaerobic conditions that lower soil Eh and increase soil pH, promoting the formation of insoluble Cd compounds, such as CdS, $CdCO_3$, and $Cd(OH)_2$. This, in turn, decreases soil Cd bioavailability (Yao et al. 2022). Conversely, a lower Eh under flooded conditions facilitates the mobilization of Hg into the soil solution and promotes its methylation by anaerobic microbes, increasing the bioavailability of both inorganic Hg and methylmercury (MeHg) to rice plants (Wang et al. 2014). This see-saw antagonistic effect poses a considerable challenge for the remediation of Cd–Hg co-contaminated paddy soils.

To enhance remediation efficacy, integrating water management with functionalized soil amendments is a promising strategy (Tian et al. 2024). Iron (Fe) and manganese (Mn) oxide-modified biochar (FMBC) has emerged as a highly effective amendment. Loading Fe and Mn oxides onto the biochar considerably optimizes its pore structure and enriches oxygen-containing functional groups (Zhou et al. 2018). These physicochemical enhancements empower FMBC to substantially reduce the bioavailability of Cd and Hg through diverse mechanisms, including surface adsorption, co-precipitation, chemical complexation, and metal–π interactions (Tang et al. 2022; Sun et al. 2023). The practical efficacy of FMBC was firmly established in our previous

Table 1 The treatments of pot experiment

Treatment	Water management	Soil amendment	Application dosage (w/w)
FCK	Continuous flooding	No amendments	0
FBC	Continuous flooding	Corn straw biochar (BC)	1%
FFM	Continuous flooding	Fe–Mn modified corn straw biochar (FMBC)	1%
ACK	Continuous aerobic	No amendments	0
ABC	Continuous aerobic	Corn straw biochar (BC)	1%
AFM	Continuous aerobic	Fe–Mn modified corn straw biochar (FMBC)	1%

studies, where its application significantly reduced Cd and Hg accumulation in rice grains by 71.16% and 72.11%, respectively (Sun et al. 2024). However, prior research has primarily evaluated the efficacy of FMBC under single-moisture conditions, failing to fully consider the impact of pronounced redox fluctuations induced by practical water management practices in paddy fields on remediation performance.

To bridge this critical gap, this study explicitly targets the inherent Cd and Hg redox antagonism by developing an integrated management strategy. By coupling the chemical immobilization capacity of FMBC with the physicochemical regulation by specific water regimes, this strategy aims to synergistically limit both Cd and Hg accumulation. We hypothesized that this synergistic remediation effectively immobilizes soil Cd and Hg by regulating soil physicochemical properties and microbial community structures. To test this hypothesis, a comprehensive pot experiment was conducted using co-contaminated paddy soil under two distinct water management regimes: continuously flooded and aerobic conditions. The overarching aim was to establish optimal water-amendment combinations and elucidate their synergistic remediation mechanisms. The specific objectives were to assess: (1) the synergistic immobilization of soil Cd and Hg; (2) the impact on rice growth and heavy metal accumulation in plants; and (3) the structural and functional responses of the soil bacterial community.

2 Materials and methods

2.1 Experimental materials and methods

Topsoil samples from 0 to 20 cm in depth were collected from a Cd and Hg co-contaminated paddy field in Tongren City (109°14' E, 27°30' N), Guizhou Province, China. The experimental soil was developed from yellow earth and was classified as hydragric paddy soil. The concentrations of Cd and Hg in polluted paddy soil were 0.76 and 5.4 mg kg^{-1}, respectively. The other physicochemical properties of the paddy soils are presented in Table S1.

Two amendments, namely, corn straw biochar (BC) and Fe–Mn modified corn straw biochar (FMBC), were used in this study. The production procedures for BC and FMBC are detailed in our previous study (Sun et al. 2024). The properties of the amendments are listed in Table S2.

2.2 Pot experiment

The pot experiment was conducted in the greenhouse of Agro-Environmental Protection Institute, MARA (39°09′07.9″N, 117°15′26″E) during May–September 2022. The experiments were conducted under natural sunlight. The average temperature during the growth period of *Oryza sativa* L. ranged from 25 to 35 °C, and relative humidity was maintained at 60–80%. The soil pot experiment included two different water management regimes throughout the rice growth period: continuously flooded (CF) and continuous aerobic (CA). For CF conditions, the water depth was maintained at 3 cm from the soil surface. For CA conditions, soil moisture was maintained at 80% of the water-holding capacity. The experiment used a randomized design comprising six treatments (two water regimes × 3 amendments), each with three replicates, totaling 18 pots (Table 1). All pots were randomly arranged in a greenhouse under natural sunlight. The experimental soils were air-dried and sieved through a 2 mm nylon mesh for the rice pot experiment. The amendments (BC and FMBC) and basal fertilizers (0.2 g N kg^{-1}, 0.3 g P kg^{-1}, 0.4 g K kg^{-1} as $CO(NH_2)_2$ and KH_2PO_4) were then thoroughly mixed into the soil before being placed into plastic pots (inner diameter, 180 mm; inner basket height, 203 mm), each containing 3.5 kg of soil. During the heading stage, supplementary fertilizers were applied at half the rate of the basal fertilizer as a water-soluble solution. Three *Indica* rice (*Oryza sativa* L. cv. Huanghuazhan) seedlings were transplanted into each pot. Pesticide application and pest control followed standard rice cultivation practices. At maturity, rhizosphere soil samples were collected using pre-cleaned stainless steel tools, homogenized, and stored in sealed polyethylene bags. Plants were harvested for biomass measurement and Cd and Hg concentration analysis.

2.3 Analysis methods

2.3.1 Soil properties analysis

Soil physicochemical properties were analyzed according to the method described by Bao (2000). Soil pH (soil:water ratio of 1: 2.5) and redox potential (Eh) were measured using a pH meter (PB-10, Sartorius, Germany) and a redox probe (Seven2Go Pro S8, Mettler Toledo), respectively. Soil cation exchange capacity (CEC) was assessed using the hexamminecobalt trichloride spectrophotometric method (HJ889-2017). Soil organic carbon (SOC) and dissolved organic carbon (DOC) were determined using the potassium dichromate method (Lu 2000).

2.3.2 Cd and Hg concentrations in soil and rice

The concentrations of Cd and total Hg (THg) in the soil and rice tissues were determined as described previously (Sun et al. 2024). The THg content in rice tissues was determined by acid digestion with 10 mL of diluted aqua regia [HNO_3:HCl:H_2O = 1:3:4 (v/v/v)] for 2 h. After cooling, the digestate was diluted to 50 mL with deionized water, filtered, and analyzed using atomic fluorescence spectrometry (AFS-8520). For Cd concentration analysis, rice tissues were digested with 10 mL of concentrated HNO_3 following a temperature-controlled protocol: 1.5 h at 80 ℃, 1.5 h at 120 ℃, and 3 h at 150 ℃, followed by acid evaporation at 170 ℃ for 0.5–1 h. The digestion process was terminated when 1–2 mL of digestate remained in the tube. The samples were diluted to 50 mL with deionized water, filtered, and analyzed using inductively coupled plasma mass spectrometry (ICP–MS, Thermo Scientific iCAP QC).

MeHg content in the rice grains was quantified using liquid chromatography–inductively coupled plasma–mass spectrometry (LC–ICP–MS) (GB 5009.17-2021). The method exhibited a limit of detection (LOD) of 0.005 mg kg^{-1} and a limit of quantification (LOQ) of 0.02 mg kg^{-1}.

A 0.01 M $Na_2S_2O_3$ solution and a DTPA (diethylenetriaminepentaacetic acid) solution (0.005 M DTPA, 0.01 M $CaCl_2$, and 0.1 M triethanolamine) were used for extracting soil available Hg ($Na_2S_2O_3$–Hg) and Cd (DTPA–Cd), respectively (Lindsay and Norvell 1978; Chang et al. 2020). The Hg and Cd fractions were determined using the sequential extraction procedure described by Tessier et al. (1979), which distinguishes five fractions: exchangeable fraction (EXC–Hg/Cd), carbonate-bound fraction (CAR–Hg/Cd), Fe/Mn oxide-bound fraction (OX–Hg/Cd), organic matter-bound fraction (OM–Hg/Cd), and residual fraction (RES).

2.3.3 Iron–manganese plaques (IMP) on root surface

IMP on the fresh root surface was extracted using DCB (dithionite–citrate–bicarbonate) solution (0.03 M $Na_3C_6H_5O_7{\cdot}2H_2O$; 0.125 M $NaHCO_3$; 1.2 g $Na_2S_2O_4$) (Taylor and Crowde 1983; Huang et al. 2022). The roots were rinsed three times with deionized water, immersed in the DCB extract solution, and diluted to 100 mL with deionized water. DCB–Fe and DCB–Mn concentrations were quantified using inductively coupled plasma mass spectrometry (iCAP Q, Thermo Scientific, USA).

2.3.4 Soil bacterial community analysis

Fresh soil samples for microbial analysis were collected using sterile spatulas, immediately placed in sterile centrifuge tubes, flash-frozen in liquid nitrogen, and transported to the laboratory on dry ice, where they were stored at − 80 ℃ until DNA extraction (Guo et al. 2025). Bacterial community sequencing was performed by the Majorbio Company in Shanghai, China. The detailed protocols for high-throughput sequencing analysis are described in Supplementary Text S1.

2.4 Data processing

The health risks of heavy metal intake among residents were assessed using acceptable daily intake (ADI) and target hazard quotient (THQ). A THQ value ≤ 1 suggests that adverse health effects are unlikely, whereas a THQ value > 1 indicates a potential risk to human health (Xu et al. 2020).

$$ADI = \frac{C_r \times M_r}{B_w} \tag{1}$$

$$THQ = \frac{ADI}{RfDo} \tag{2}$$

where C_r is the concentration of Cd/Hg in rice grains (mg kg^{-1}); M_r represents the rice intake rate (g d^{-1}), which, according to the Chinese Dietary Guidelines for Food and Nutrition Development, it is set at 370 g d^{-1}; and B_w denotes the body weight of residents (kg), with a standard body weight established at 61.7 kg. The oral reference dose (*RfDo*) refers to the daily intake limits for Cd, THg, and MeHg recommended by the USEPA, which are 0.57, 0.1, and 1 μg kg^{-1} d^{-1}, respectively.

2.5 Quality control and statistical analysis

Quality control was rigorously implemented throughout the analytical procedure, including replicate analyses, sample blanks, and standard reference materials. The soil standard reference material SRM 2586 (National Institute of Standards and Technology, NIST) was used to monitor

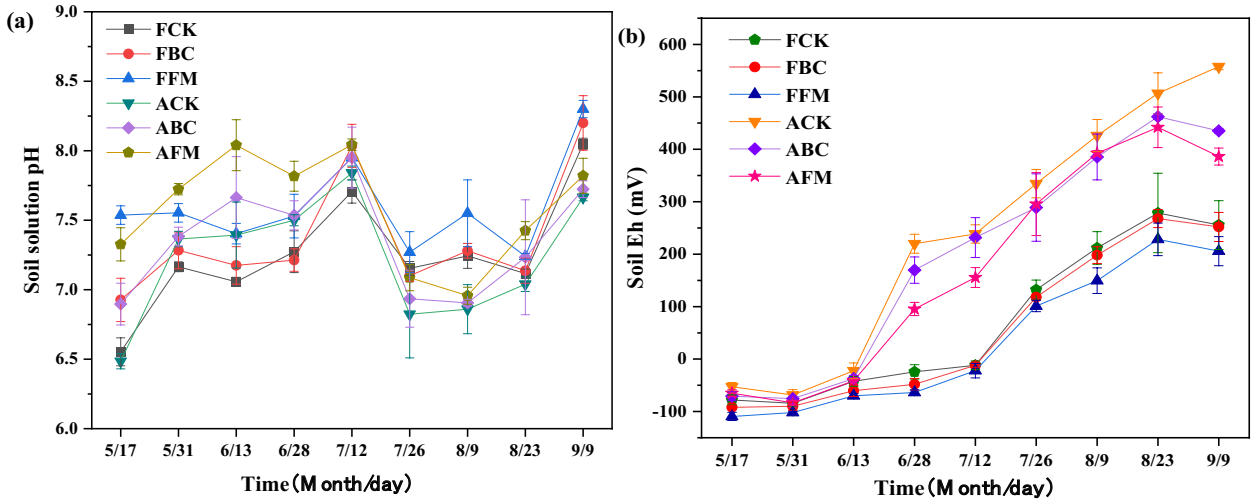


Fig. 1 Effects of FMBC combined with water management on soil solution pH (**a**) and Eh (**b**) at different rice growth stages

digestion and determination (recovery 89.4–105.7%). Rice flour standard reference materials (GBW 100561 and GBW 100349) were used for quality control in rice Cd and THg determination (recovery 94.6–102.8%). The standard material ERM-CC-580 was used for quality control in rice MeHg determination (recovery 91.3–108.4%).

Statistical analysis and figure drawing were conducted using DPS 9.01 and Origin 2021 software. Data are presented as mean ± standard deviation ($n=3$). Prior to ANOVA, normality of residuals was assessed using the Shapiro–Wilk test, and homogeneity of variances was verified using Levene's test. All data met the assumptions of normality and homoscedasticity ($P>0.05$). Two-way ANOVA was used to evaluate the effects of water management, amendment, and their interaction on Hg and Cd concentrations in rice tissues. When a significant interaction was detected ($P<0.05$), simple main effects analysis was subsequently performed. One-way ANOVA was employed to compare the effects of different treatments on soil physicochemical properties, bioavailable Hg/Cd concentrations, rice biomass, IMP, and microbial community parameters. Post-hoc multiple comparisons were performed using Duncan's multiple range test, with statistical significance set at $P<0.05$. The partial least squares path modeling (PLSPM) analysis was conducted using the *plspm* package in R software (version 4.2.2). The model goodness-of-fit (GoF) was evaluated via cross-validation.

3 Results and discussion

3.1 Soil properties and metal bioavailability

3.1.1 Soil physical and chemical characteristics

Water management regimes influence the bioavailability of heavy metals by altering key soil physicochemical properties, such as pH, Eh, CEC, and SOC, which in turn influence their accumulation in rice tissues (Xia et al. 2024). The soil solution pH exhibited significant fluctuations throughout the rice growth stages, generally demonstrating an initial increase, followed by a decline, and then a subsequent increase (Fig. 1a). Studies have shown that under continuous flooding conditions, soil pH gradually converges toward neutral levels, irrespective of the initial soil acidity or alkalinity (Pan et al. 2016). In this study, during the early growth stage of rice, the application of BC and FMBC under flooded conditions led to a higher soil solution pH than aerobic treatments. At harvest, the pH of the flooded soil remained elevated by 0.03–0.09 units relative to aerobic conditions. This may be due to the soil undergoing a reduction reaction and consuming a large number of protons after long-term flooding, resulting in a higher soil pH (Yao et al. 2022). Both BC and FMBC applications significantly enhanced soil pH compared to the control, consistent with their inherent alkaline properties.

Soil Eh is a critical indicator of the redox status in paddy soils (Liu et al. 2022). Our results showed that soil Eh is strongly influenced by water management practices. Soil Eh was significantly lower in the flooded treatments than in the aerobic treatments and showed an increasing trend over the experimental period (Fig. 1b). The range of soil Eh values varied between −109.42 and 278.60 mV for flooded treatments and between −70.76 and 557.67 mV for aerobic treatments. Long-term flooding irrigation causes a decline in oxygen content in paddy soil. Anaerobic microorganisms decompose organic matter, producing large amounts of reducing substances, which lower soil Eh and result in a reduced state in the paddy soil (Li et al. 2022a). BC and FMBC addition under both water

management regimes significantly reduced soil Eh values compared with the control. This could be attributed to the unstable organic compounds in BC and FMBC, which enhanced microbial respiration and accelerated soil O_2 consumption, increasing the relative quantity of reducing substances (Yin et al. 2017).

The different water management regimes had no significant influence on soil CEC, SOC, and DOC contents (Fig. S1). However, BC and FMBC applications enhanced the soil SOC and DOC contents under both water management regimes relative to the control. BC was more effective than FMBC, with SOC and DOC contents increasing by 30.65% and 19.07% under flooded conditions and by 37.55% and 8.09% under aerobic conditions, respectively ($P < 0.05$). These findings align with those of a previous study (Jiang et al. 2022), in which BC and FMBC contained substantial amounts of recalcitrant condensed aromatic structures that contributed to the stabilization and accumulation of SOC.

3.1.2 Soil Cd and Hg bioavailability and fractions

Different water management regimes had significantly opposite effects on the available Cd and Hg contents in the soil (Fig. 2a and b). The soil available Cd (DTPA–Cd) content under continuous flooding treatment was significantly higher than that under continuous aerobic treatment. The FCK treatment resulted in a 3.47% reduction in the DTPA–Cd concentration relative to the ACK treatment ($P < 0.05$). Continuous flooding irrigation results in the soil remaining in a reduced state (Eh decreases) and enhances the growth of anaerobic microorganisms, such as sulfate-reducing bacteria. SO_4^{2-} in the soil is reduced to S^{2-}, which is highly likely to combine with Cd^{2+} to form insoluble CdS (Tian et al. 2019; Xu et al. 2021). While under aerobic conditions, the oxidation of CdS to Cd^{2+} and SO_4^{2-} can enhance the concentration of plant-available Cd (Furuya et al. 2016).

For Hg, the $Na_2S_2O_3$–Hg content in continuous aerobic treatments was significantly reduced by 20.66–22.29% ($P < 0.05$), compared with the continuous flooding treatments. The Hg fraction was predominantly residual (RES–Hg), accounting for 83.58–85.96% and 82.00–85.93% in flooded and aerobic treatments, respectively (Fig. 2d). The bioavailability of Hg is closely associated with soil Eh (Delaune et al. 2004). Peng et al. (2012) reported significantly higher soil Eh under aerobic conditions than under flooded conditions. Concurrently, the contents of THg and MeHg in soil solution were approximately 2 and 4 times greater, respectively, under continuous flooding than in aerobic regimes, indicating a heightened risk of mercury accumulation under continuous flooding conditions.

Compared to the control, the application of BC and FMBC further reduced the contents of DTPA–Cd and $Na_2S_2O_3$–Hg in soil under both flooded and aerobic conditions ($P < 0.05$). FMBC demonstrated greater bioavailability reduction efficiency, decreasing DTPA–Cd by 1.81% (flooded) and 3.41% (aerobic), and $Na_2S_2O_3$–Hg by 27.84% (flooded) and 26.86% (aerobic). Compared with the no-amendment treatments, FMBC significantly reduced the percentages of EXC–Cd and CAR–Cd while increasing the percentages of OX–Cd, OM–Cd, and RES–Cd (Fig. 2c). This fraction transformation suppressed the bioavailability of Cd and Hg in the soil system, consistent with our previous findings (Sun et al. 2024). The enhanced stabilization capacity of FMBC was attributed to its elevated pH and larger specific surface area. This facilitated key physicochemical interactions such as electrostatic adsorption, cation exchange, precipitation, complexation, and π–metal interactions with Cd and Hg.

3.2 Rice yield and iron–manganese plaques on rice root surface

3.2.1 Rice growth and biomass

Flooded conditions can enhance crop yields compared to aerobic treatments (Carrijo et al. 2017; Wu et al. 2017). Meanwhile, other studies have found no significant effects of water management on crop productivity (Grigg et al. 2000). Our data showed that the continuously flooded treatment significantly enhanced rice growth and yield (Table 2). Brown rice yield significantly increased by 12.86–24.14%, compared to the aerobic treatment ($P < 0.05$). This may be related to the buildup of soil pathogens and nematodes under aerobic conditions (Hu et al. 2015). BC and FMBC applications promoted rice growth and tillering under both water management regimes. The yield increased in brown rice, reaching 14.02% and 30.98% under flooded and aerobic conditions, respectively.

3.2.2 Iron–manganese plaques (IMP) on rice root surface

The formation of IMP on rice root surfaces was affected by water management regimes (Fig. S2). Flooding consistently promoted IMP development, resulting in DCB–Fe and DCB–Mn concentrations that were 6.49–12.05 and 2.81–5.34 times higher, respectively, under flooded conditions than under aerobic treatments. IMP on the rice root surface exhibits a strong Cd-binding affinity, and acts as a critical interface regulating Cd transport from soil to roots (Cheng et al. 2014; Zhang et al. 2019). In this study, continuously flooded treatment promoted the formation of IMP on the rice root surface compared to the aerobic treatment, consistent with previous research (Xiao et al. 2021). Huang et al. (2019) found that flooding

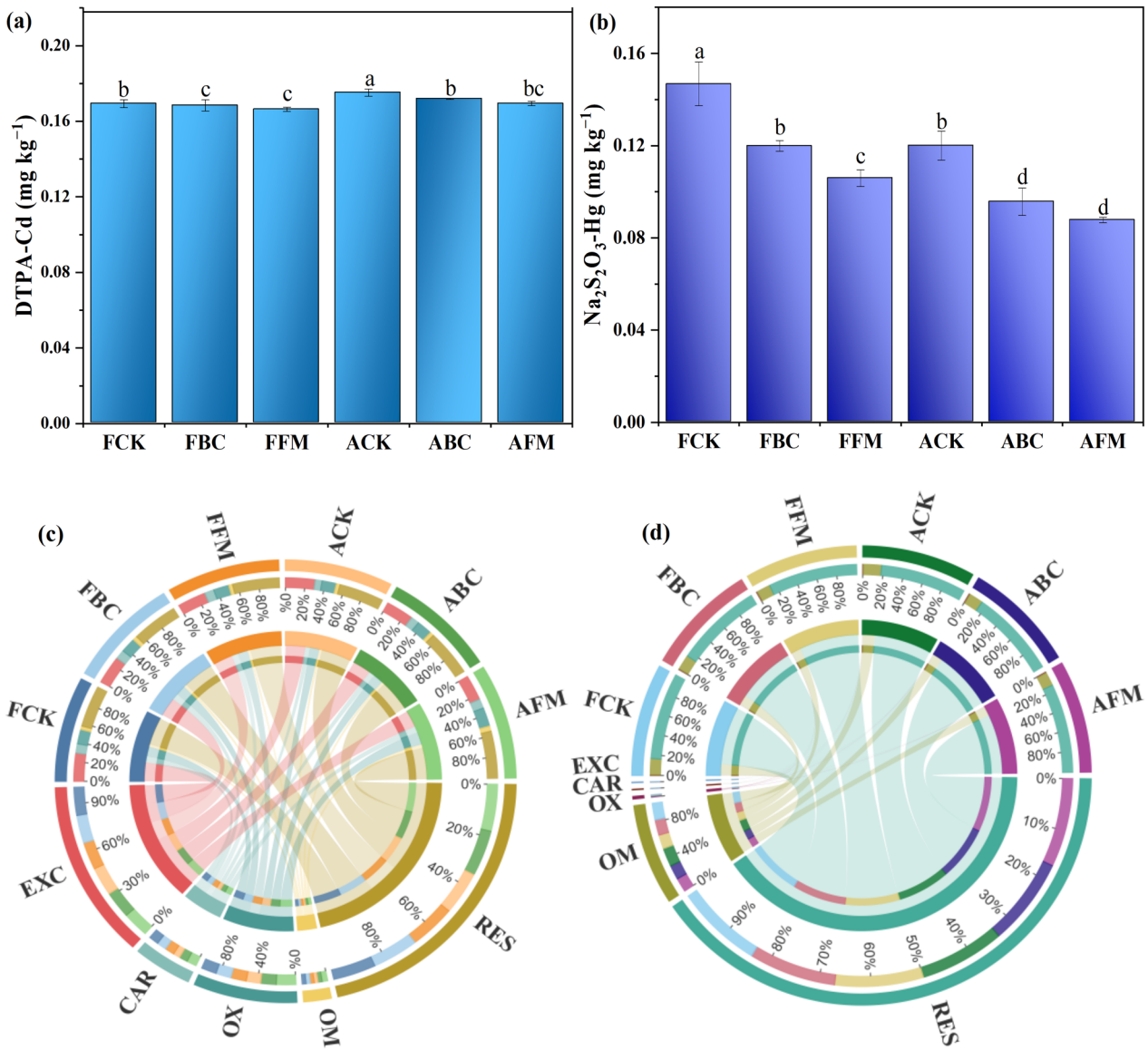


Fig. 2 Effects of FMBC application and water regimes on the concentration of DTPA–Cd (**a**), $Na_2S_2O_3$–Hg (**b**), Cd fractions (**c**), and Hg fractions (**d**) in soils. Error bars represent SD of three replicates. Different letters indicate significant difference ($P < 0.05$) from each other according to the Duncan's test

Table 2 Effects of FMBC application and water regimes on rice growth and biomass

Treatment	Plant height (cm)	Spike number	Dry weights (g/plant)				
			Root	Stem	Leaf	Grain	Panicle rachis
FCK	107.00 ± 1.73 c	8.50 ± 0.17 a	2.45 ± 0.38 b	4.68 ± 0.68 c	12.68 ± 1.31 b	18.53 ± 0.84 ab	1.07 ± 0.07 ab
FBC	112.67 ± 4.51 ab	9.33 ± 0.33 a	3.67 ± 0.36 a	6.99 ± 1.55 ab	15.06 ± 1.65 a	20.29 ± 1.98 ab	1.02 ± 0.13 ab
FFM	113.50 ± 3.77 a	8.78 ± 1.64 a	3.41 ± 0.68 a	7.06 ± 0.77 ab	15.46 ± 1.27 a	21.13 ± 3.11 a	1.13 ± 0.15 a
ACK	101.67 ± 2.08 d	8.83 ± 0.50 a	1.67 ± 0.36 c	5.79 ± 0.74 bc	14.10 ± 0.92 ab	14.06 ± 2.07 c	0.94 ± 0.03 ab
ABC	108.00 ± 1.00bc	9.50 ± 0.50 a	3.25 ± 0.33 ab	7.88 ± 0.42 a	15.89 ± 0.04 a	15.49 ± 2.79 bc	0.96 ± 0.10 ab
AFM	107.33 ± 0.58 c	9.17 ± 0.17 a	3.05 ± 0.38 ab	5.25 ± 0.79 c	15.85 ± 0.66 a	18.41 ± 0.26 ab	0.85 ± 0.05 b

the soil can significantly enhance the Fe and Mn contents in the soil solution relative to aerobic treatments, stimulating IMP formation. Continuous flooding induces an anaerobic environment that drives the reductive dissolution of Fe^{3+} into Fe^{2+}. This process lowers soil Eh and supplies Fe^{2+} for the formation of IMP on rice roots, which can sequester Cd. Furthermore, to adapt to the anaerobic environment of long-term flooding, aquatic plants develop extensive aerenchyma in their roots, and the oxygen transported by rice tissues is released into the roots and rhizosphere. Fe and Mn ions in the soil solution undergo oxidation and react with hydroxides to form iron–manganese oxide precipitates on the rice root surfaces (Kong et al. 2024).

The FMBC application promoted IMP formation on the root surface under both water management regimes. This may be due to the addition of FMBC, which increases the soluble iron and manganese ions near the rice roots, facilitating the formation of iron–manganese oxides (Wang et al. 2021a). Under continuous aerobic conditions, both BC and FMBC significantly enhanced the DCB–Fe concentration in rice roots by 82.93% and 64.98%, respectively, relative to the control ($P < 0.05$). However, no significant difference was observed between these two treatments ($P > 0.05$). Under the continuous flooding conditions, only the FFM treatment demonstrated a 10.85% increase in DCB–Fe content compared with the FCK treatment ($P < 0.05$). The addition of FMBC significantly enhanced the DCB–Mn content under both water management conditions. Compared with the control, the DCB–Mn content was enhanced by 4.33 and 8.33 times under the continuous aerobic and flooded conditions, respectively ($P < 0.05$).

3.3 Cd, THg, and MeHg accumulation in rice and its health risk assessment

3.3.1 Cd uptake and accumulation

As shown in Fig. 3, the Cd concentration in rice tissues was significantly higher under continuous aerobic conditions compared to continuous flooding conditions. Under no amendment treatment, Cd concentration in rice grains under the aerobic condition was approximately 4.00–8.49 times higher than that under the flooded treatment. Similar conclusions have been drawn in other studies. Xu et al. (2014) found that compared to flooded irrigation (FI), non-flooded irrigation (NFI) resulted in higher uptake of Cr and Cd by rice plants and could potentially pose food safety risks in a short period. Li et al. (2017) demonstrated that under continuous flooding management, the contents of soil-exchangeable Cd and brown rice Cd were lower than those under traditional and wetting irrigation management, which helped implement soil Cd immobilization with palygorskite. Prolonged soil flooding led to decreases in soil Eh and SO_4^{2-}, promoting the formation of reducing substances. This shift enhanced the growth of anaerobic microorganisms and decreased the bioavailability of Cd, affecting Cd accumulation in rice tissues (Li et al. 2021a).

Previous studies have demonstrated that combining water management practices with soil amendments, such as biochar (Cang et al. 2020), calcium-magnesium-phosphorus fertilizers (Luo et al. 2020), and sepiolite (Zhu et al. 2012), can significantly inhibit heavy metal accumulation in crops. In this study, the FMBC application significantly reduced the Cd content in rice tissues under both flooded and aerobic irrigation conditions. Compared with CK, the concentrations of Cd in the roots, stems, leaves, and grains were reduced by 56.08%, 72.65%, 68.30%, and 72.62% under the continuously flooded treatments, respectively, and reduced by 67.72%, 64.64%, 61.15%, and 41.92% under the continuous aerobic treatment, respectively ($P < 0.05$). Under FFM treatment, the Cd concentration in the grains reached the lowest value (0.05 mg kg^{-1}), which was significantly below the national limit of 0.2 mg kg^{-1} (GB 2762-2017). Two-way ANOVA indicated a highly significant interaction between water management and FMBC addition on grain Cd accumulation ($P < 0.01$). Continuous flooding induces aerenchyma development and radial oxygen loss in rice roots as an adaptation to anaerobic stress. This biological oxidation process interacts with the abundant soluble Fe and Mn ions released by FMBC, thereby accelerating the IMP formation on the root surface (Zeng et al. 2022). Consequently, the FFM treatment formed a more robust physiological barrier against Cd uptake than the additive effects of single factors.

Under the continuous aerobic treatments, the ADI of Cd ranged from 2.61 to 4.49 μg kg^{-1} d^{-1} (Table 3), well above the permissible limit of 1 μg kg^{-1} d^{-1} recommended by the U.S. Environmental Protection Agency (USEPA), indicating a significant non-carcinogenic risk of Cd exposure through rice consumption. Under the continuous flooding conditions, the THQ of Cd in grain in the FCK treatment was 1.23 (THQ > 1), indicating a higher consumption risk for residents. However, BC and FMBC treatments significantly reduced THQ values to 0.65 and 0.31, respectively, thereby decreasing the potential health risks to residents ($P < 0.05$). In summary, the combination of flooded irrigation and FMBC application is an effective technique for minimizing the risks of Cd exposure from rice consumption.

3.3.2 THg and MeHg uptake and accumulation

Contrary to the Cd results, when comparing the THg content in rice roots, stems, leaves, grains, rachises, and husks under the two water management regimes, the

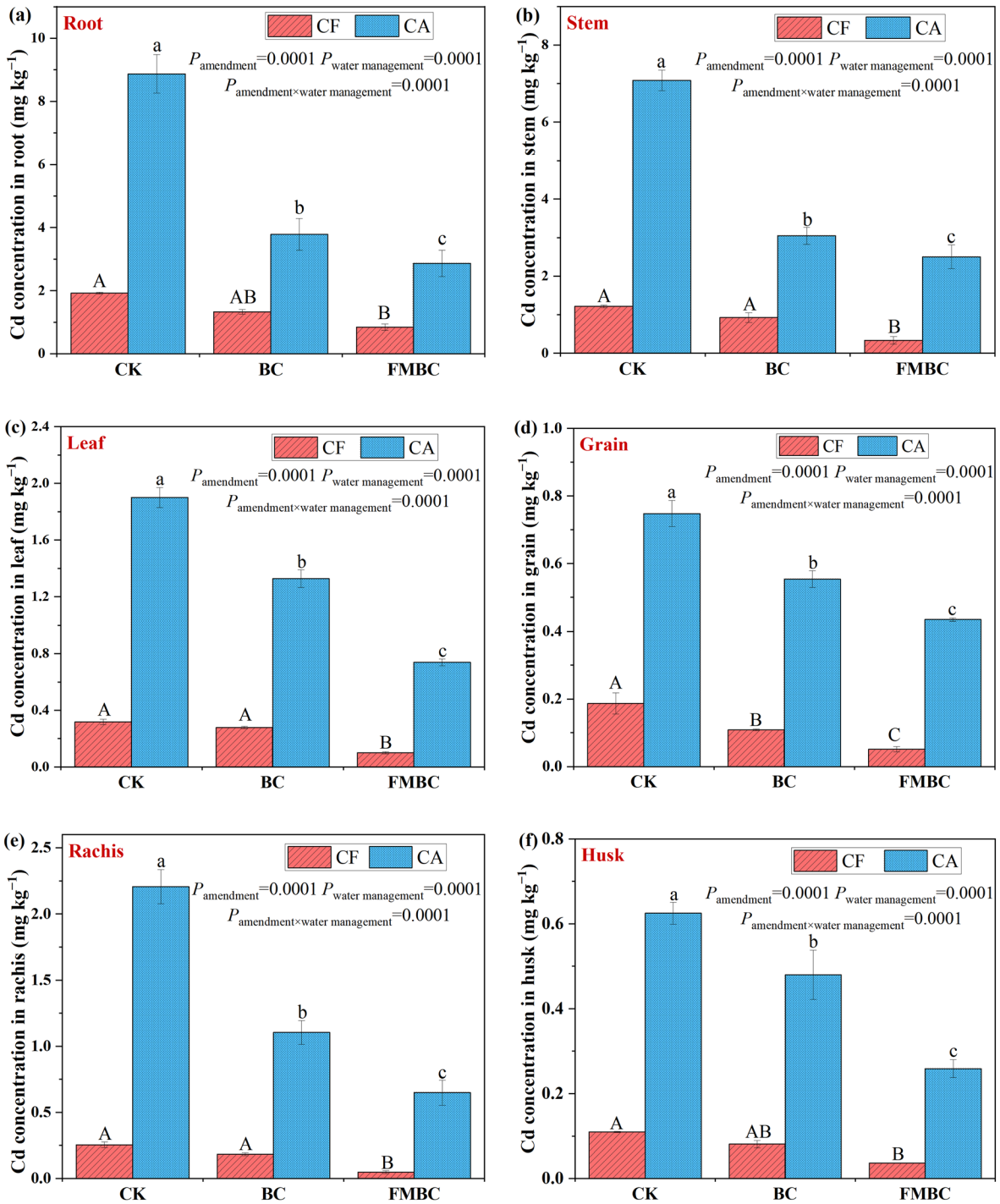


Fig. 3 Influences of FMBC combined with water management on Cd concentration in rice tissues of **a** root, **b** stem, **c** leaf, **d** grain, **e** rachis, and **f** husk. Different letters indicate significant differences among treatments ($P < 0.05$). Uppercase and lowercase letters denote differences under continuous flooded (CF) and continuous aerobic (CA) conditions, respectively

THg concentration in various rice tissues under continuous aerobic conditions was substantially lower than that under the continuous flooding treatment (Fig. 4a–f). The THg content in grains under the continuous flooding treatment (0.05–0.10 mg kg^{-1}) was 2.55–3.46 times higher than that under the aerobic treatments (0.02–0.03 mg kg^{-1}). A similar trend was observed for MeHg (Fig. 4g), with its content in rice grains being significantly

Table 3 Effect of different treatments on the health risk assessment of THg, MeHg and Cd in rice grain

Treatment	Cd		THg		MeHg	
	ADI (μg kg^{-1} d^{-1})	THQ	ADI (μg kg^{-1} d^{-1})	THQ	ADI (μg kg^{-1} d^{-1})	THQ
FCK	1.23 ± 0.02 d	1.23 ± 0.02 d	0.65 ± 0.04 a	1.15 ± 0.08 q	0.42 ± 0.02 a	4.29 ± 0.12 a
FBC	0.65 ± 0.02 e	0.65 ± 0.02 e	0.50 ± 0.05 b	0.84 ± 0.04 b	0.32 ± 0.04 b	3.42 ± 0.02 b
FFM	0.31 ± 0.05 f	0.31 ± 0.05 f	0.28 ± 0.04 c	0.46 ± 0.04 c	0.23 ± 0.01 c	2.28 ± 0.07 c
ACK	4.49 ± 0.23 a	4.49 ± 0.23 a	0.19 ± 0.01 d	0.33 ± 0.02 d	0.07 ± 0.01 d	0.75 ± 0.03 d
ABC	3.32 ± 0.15 b	3.32 ± 0.15 b	0.16 ± 0.01 d	0.28 ± 0.01 de	0.06 ± 0.01 d	0.64 ± 0.01 e
AFM	2.61 ± 0.03 c	2.61 ± 0.03 c	0.13 ± 0.02 d	0.22 ± 0.03 e	0.04 ± 0.01 d	0.41 ± 0.01 f

higher under continuous flooding than under the aerobic treatment. Under the no amendment treatments, the MeHg content in the continuous aerobic treatment decreased by 82.93% compared to that in the continuous flooding treatment. Furthermore, the proportion of THg existing as MeHg in grains dropped from 64.82 to 69.85% under flooded conditions to 32.68–33.95% under aerobic conditions (Fig. 4h). These findings align with previous reports that aerobically grown rice exhibits markedly lower Hg and MeHg accumulation (Peng et al. 2012). Hg methylation is a microbially mediated process strongly regulated by soil Eh (Ran et al. 2022; Xing et al. 2022). Long-term flood irrigation alters soil aeration and redox processes, significantly lowering soil Eh values, creating favorable conditions for Hg methylation. As shown in the subsequent microbial community analysis (Sect. 3.4.3), this low-Eh environment promoted the growth and enrichment of *hgcA*-containing anaerobic microorganisms. The enhanced activity of these methylators drives MeHg production in the soil, leading to increased Hg and MeHg bioavailability and accumulation in rice plants (Wang et al. 2014).

Relative to the no-amendment treatments, both BC and FMBC amendments significantly reduced the THg and MeHg contents in rice plant tissues. FMBC was more effective in suppressing Hg accumulation in rice plants than BC under identical water management practices. Under continuous flooding, FMBC addition reduced THg concentrations in roots, stems, leaves, and grains by 22.57%, 21.26%, 49.92%, and 72.62%, respectively ($P < 0.05$). Corresponding reductions under aerobic conditions were 28.49%, 29.38%, 25.35%, and 32.04% ($P < 0.05$). Notably, aerobically produced grains treated with FMBC had the lowest THg concentration (0.02 mg kg^{-1}), reaching the national standard limit (GB 2762-2017). Furthermore, FMBC application significantly decreased grain MeHg content by 44.59% and 38.83% under flooded and aerobic conditions, respectively ($P < 0.05$). Statistical analysis revealed a highly significant interaction effect of water management and FMBC addition on THg and MeHg accumulation ($P < 0.01$). Fundamentally, soil MeHg production is primarily determined by the concurrent activity of Hg-methylating bacteria and the bioavailability of inorganic Hg (IHg). Suppressing IHg bioavailability can effectively inhibit methylation because only a small, accessible fraction of Hg is utilized for this process (Man et al. 2021). In this study, aerobic irrigation biologically suppressed the relative abundance of anaerobic *hgcA*-containing microbes. Concurrently, the specific surface functional groups of FMBC—such as oxygen-containing groups, metal-oxygen bonds, and sulfur-containing compounds—chemically complex and immobilize the inorganic Hg (IHg) substrate (Sun et al. 2024). By concurrently restricting IHg bioavailability and suppressing the total abundance of *hgcA*-containing methylators (Fig. 5e), this combined treatment synergistically curtails both direct Hg uptake and MeHg production, minimizing grain THg and MeHg accumulation far more effectively than single factors.

The health risks of THg and MeHg intake were assessed using THQ and ADI (Table 3). Under flooded treatments, THQ values for both THg and MeHg exceeded 1, indicating potential health risks. Conversely, under continuous aerobic treatments with FMBC, the ADIs of THg and MeHg were 0.13–0.19 and 0.04–0.07 μg kg^{-1} d^{-1}, respectively, which were safely below the USEPA limits (0.57 and 0.1 μg kg^{-1} d^{-1}). Ultimately, the AFM treatment achieved the lowest THQ levels for THg (0.22) and MeHg (0.41), confirming that this specific integration effectively mitigates dietary Hg exposure risks.

3.4 Soil bacterial community diversity and function

3.4.1 Sequence data and bacterial community diversity

Across all soil samples, we obtained 733,837 high-quality bacterial V4–V5 Illumina HiSeq sequencing reads after applying all quality filters with a mean of 40,768 sequences per sample. The analysis showed 1578 core operational taxonomic units (OTUs) shared among all soils, representing 62.10% of the total number of OTUs (Fig. 5a). The numbers of unique OTUs in FCK, FBC,

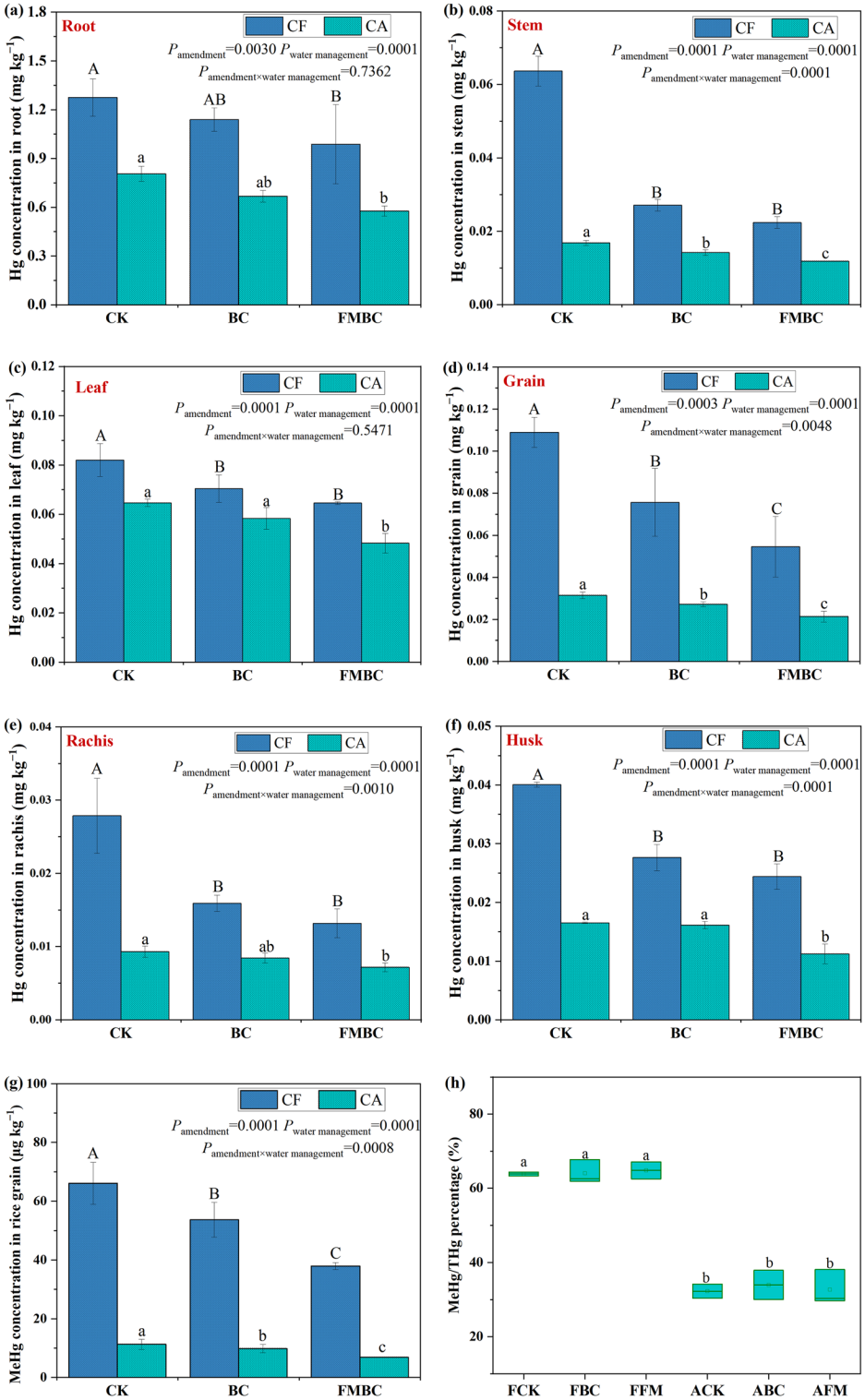


Fig. 4 Effects of FMBC combined with water management on Hg content in **a** root, **b** stem, **c** leaf, **d** grain, **e** rachis and **f** husk, and **g** MeHg in grain, **h** MeHg/THg percentage in grain. Different letters indicate significant differences among treatments ($P < 0.05$). Uppercase and lowercase letters denote differences under continuous flooded (CF) and continuous aerobic (CA) conditions, respectively

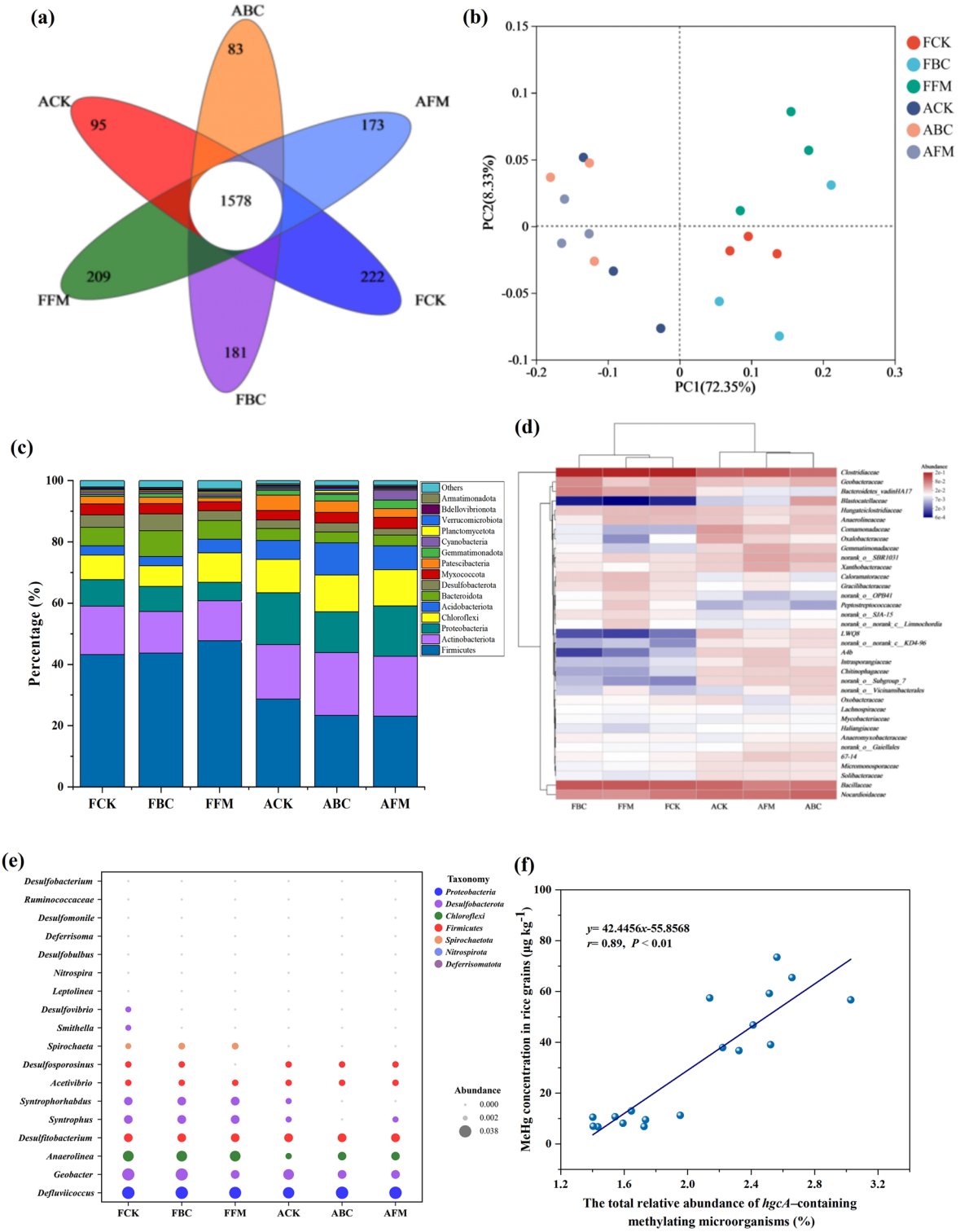


Fig. 5 Venn diagram showing differences in microbial community compositions (**a**); PCoA exhibiting the differences among water management regimes (**b**); Relative abundance of bacterial composition at phyla level under water management regimes (**c**); Heatmap of dominant bacterial genera under different water management regimes (**d**); The relative abundance (%) of bacteria containing *hgcA* gene sequence at genus level (**e**); Linear relationship between the MeHg concentration in rice grains and the total relative abundance of *hgcA*-containing methylating microorganisms (**f**)

FFM, ACK, ABC, and AFM were 221, 181, 209, 95, 83, and 173, respectively. The flooded treatments contained significantly more unique OTUs than the aerobic treatments, suggesting that water management regimes substantially altered the number of species in the soil bacterial community.

Water management regimes have essential effects on the soil bacterial community structure, diversity, and function (Nie et al. 2025). In this study, the sequencing coverage reached 99.5% for all samples, which confirms that the sequencing depth accurately represents the true situation of bacterial communities (Table 4). Both water management and soil amendments alter the soil environment, resulting in alterations in the diversity and composition of the bacterial community. Compared with continuous aerobic conditions, continuous flooding significantly increased the Simpson, ACE, and Chao1 indices ($P < 0.05$) and decreased the Shannon index. This indicated that flooding was beneficial to the richness of the bacterial community but not conducive to its diversity. This may be because prolonged flooding substantially reduces oxygen levels in the soil, making it difficult for most aerobic bacteria to survive for extended periods and resulting in a decrease in bacterial community diversity. Compared with unamended soils, the addition of BC and FMBC significantly improved the diversity and richness of the soil bacterial community under both continuous aerobic and flooding conditions. This is because BC and FMBC improve soil aeration and water retention, promoting microbial growth and reproduction (Yao et al. 2017).

3.4.2 Bacterial community structure and composition

According to the hierarchically clustered heat map of the bacterial community at the family level (Fig. 5d), the soil samples were clustered into two distinct clusters. FCK, FBC, and FFM were separated from ACK, ABC, and AFM, indicating that water management significantly influenced the bacterial community structure. To visualize treatment-specific differences in soil microbial communities, PCoA analysis of the bacterial community composition was performed using the weighted UniFrac algorithm (Fig. 5b). The first two principal coordinates accounted for 72.32% (PC1) and 8.33% (PC2) of the total variance across six treatments. PCoA showed distinct clustering of flooded versus aerobic treatments, indicating that water management regimes had stronger effects on bacterial community composition than amendment applications (BC and FMBC).

Water management regimes significantly influenced the relative abundance of the dominant bacterial phyla (Fig. 5c). Across all treatments, the phylum-level community was dominated by Firmicutes, Actinobacteria, Proteobacteria, Chloroflexi, Acidobacteria, and Bacteroidetes. Firmicutes were the most abundant phylum observed in the soil samples, with a relative abundance of 44.65–47.77% and 23.12–28.67% under flooded and aerobic irrigation regimes, respectively. Firmicutes are anaerobic or facultatively anaerobic microorganisms capable of surviving and reproducing in anaerobic environments. Continuously flooded irrigation significantly enhanced the relative abundance of Firmicutes compared to aerobic treatments. Conversely, aerobic treatments increased the relative abundance of Actinobacteria and Proteobacteria by 1.92–6.82% and 5.16–10.42% ($P < 0.05$), respectively, compared to flooded conditions. Actinobacteria belong to oligotrophic groups, and the low availability of carbon sources in the soil under aerobic conditions stimulates their survival and proliferation, resulting in a relatively higher abundance in the continuous aerobic treatments (Yan et al. 2021). Proteobacteria exhibit strong dominance during drought stress (Bi et al. 2020).

To further investigate how water management and soil amendments affect microbially mediated biochemical processes, we analyzed the variations in key bacterial families. The dominant families observed across all the treatments included Clostridiaceae, Bacillaceae, Nocardioidaceae, Geobacteraceae, and Bacteroidetes (Fig. 5d). Flooded irrigation significantly enhanced the abundance of Clostridiaceae and Bacillaceae compared with the aerobic treatment. The relative abundances of Clostridiaceae and Bacillaceae were 18.53–18.75% and 8.02–9.18% under the flooded treatment, respectively. Meanwhile, under aerobic treatment, their relative abundances were only 6.45–8.31% and 4.58–7.20%, respectively. The family Clostridiaceae is predominantly found in soil, particularly in anaerobic or microaerophilic environments, and plays a key role in Hg methylation (He et al. 2023). In the present study, Clostridiaceae exhibited a higher relative abundance under continuous flooding treatment. The Bacillaceae family belongs to a group of gram-positive bacteria capable of producing resistant endospores and their cell walls contain negatively charged groups that can electrostatically interact with heavy metal ions, facilitating their adsorption and immobilization. Strains of the genus *Bacillus* within the Bacillaceae family exhibit a strong tolerance to various heavy metal ions (Wang et al. 2019a; Yang et al. 2024). Furthermore, compared to the aerobic treatment, flooded conditions significantly enhanced the relative abundance of Geobacteraceae. As a typical family of iron-reducing bacteria (FeRB) that thrives exclusively in strictly anaerobic environments, Geobacteraceae strains exhibit a significant capacity for mercury methylation (Wang et al. 2019b; He et al. 2023). This indicates that flooding irrigation promotes the growth of FeRB, which aligns with the findings of Xiao

Table 4 Effects of FMBC combined with water management regimes on alpha diversity index of soil bacteria

Treatment	Community diversity		Community richness		Coverage
	Shannon	Simpson	ACE	Chao 1	
FCK	6.086 ± 0.023 d	0.010 ± 0.001 b	3599.86 ± 45.96 a	3544.31 ± 52.88 a	0.967
FBC	6.189 ± 0.081 bcd	0.011 ± 0.001 a	3341.93 ± 129.01 b	3456.22 ± 81.13 a	0.967
FFM	6.132 ± 0.055 cd	0.010 ± 0.001 b	3463.05 ± 14.27 b	3504.56 ± 35.58 a	0.968
ACK	6.212 ± 0.004 bc	0.007 ± 0.001 c	2777.48 ± 84.22 d	2790.30 ± 96.16 c	0.973
ABC	6.280 ± 0.001 ab	0.006 ± 0.001 d	2855.76 ± 30.64 d	2858.45 ± 95.65 c	0.973
AFM	6.358 ± 0.032 a	0.005 ± 0.001 d	3024.60 ± 83.24 c	3042.25 ± 94.51 b	0.972

et al. (2022), who reported that continuous flooding stimulates the growth of sulfate-reducing and iron-reducing bacteria in paddy soils.

3.4.3 Methylating microorganisms containing hgcA gene sequence

The *hgcA* gene is a characteristic gene found in many methylating microorganisms, and its presence is strongly correlated with microbial methylation ability (Wang et al. 2024). Microbes that possess the gene cluster necessary for Hg methylation (*hgcA*) are predominantly anaerobic, encompassing groups such as iron-reducing bacteria, sulfate-reducing bacteria, and methanogens (Huang et al. 2024). Figure 5e shows the relative abundance of soil bacterial genera that possess the *hgcA* sequence after rice harvesting under different water regimes. In all six treatments, *Defluviicoccus*, *Geobacter*, *Anaerolinea*, *Desulfitobacterium*, *Syntrophus*, and *Syntrophorhabdus* were the dominant genera in *hgcA*-containing bacterial communities. These genera are primarily distributed across the phyla Proteobacteria, Chloroflexi, Firmicutes, and Desulfobacterota. The total relative abundance of *hgcA*-containing methylating microorganisms under continuous flooding and aerobic treatments was 2.34–2.58% and 1.51–1.78%, respectively. The continuous flooding treatment exhibited significantly higher relative abundances of *Defluviicoccus*, *Geobacter*, *Anaerolinea*, *Syntrophus*, *Syntrophorhabdus*, *Desulfovibrio*, and *Spirochaeta* compared to the aerobic treatment. This anaerobic microhabitat preference aligns with prior evidence linking Hg methylation potential to reduced redox conditions (Rothenberg et al. 2016; Liu et al. 2023a).

Under continuous flooding conditions, the addition of FMBC reduced the relative abundance of *Geobacter*, *Desulfosporosinus*, *Desulfovibrio*, and enhanced the relative abundance of *Defluviicoccus*, *Anaerolinea*, *Desulfitobacterium*, and *Spirochaeta* under continuous flooding conditions, compared with the control treatment. However, under continuous aerobic conditions, the addition of FMBC decreased the relative abundances of *Geobacter*, *Syntrophorhabdus*, and *Syntrophus*, while increasing those of *Defluviicoccus*, *Anaerolinea*, *Desulfitobacterium*, and *Desulfovibrio* compared to the control. The FMBC application reduced the total relative abundance of *hgcA*-containing methylating microorganisms under both the continuous flooding and aerobic irrigation regimes. Specifically, the relative abundance of *hgcA*-containing bacteria was reduced by 0.22% (FFM treatment) and 0.26% (AFM treatment) relative to the control, reducing the risk of Hg methylation in soil. Our findings show an apparent contradiction. While a lower Eh is generally recognized to enhance Hg mobility and methylation (Wang et al. 2014; Zhao et al. 2020), the addition of FMBC in this study further decreased soil Eh (Fig. 1b) yet suppressed the relative abundance of *hgcA*-containing methylating microorganisms and reduced MeHg accumulation in rice grains. This unexpected outcome may be attributed to the multifaceted regulatory mechanisms underlying FMBC. The oxygen-containing functional groups, metal-oxygen bonds, and sulfur-rich compounds on the FMBC surface effectively complex with and immobilize IHg in the soil, lowering its bioavailable pool and limiting the substrate available for microbial methylation (Wang et al. 2019c; Tian et al. 2021). In contrast, FMBC promotes the generation of reducing substances via its own reductive components while adsorbing and immobilizing reactive Hg, reshaping Hg speciation pathways at the micro-environmental scale (Xing et al. 2022).

3.4.4 Co-occurrence network analysis of the bacterial community

To evaluate how water management and FMBC incorporation influenced the interactions among bacterial taxa across abundance levels, genus level co-occurrence networks were constructed (Fig. 6), and their main topological characteristics were illustrated (Table S3). Most of the nodes in the co-occurrence network of each treatment belonged to Firmicutes, Proteobacteria,

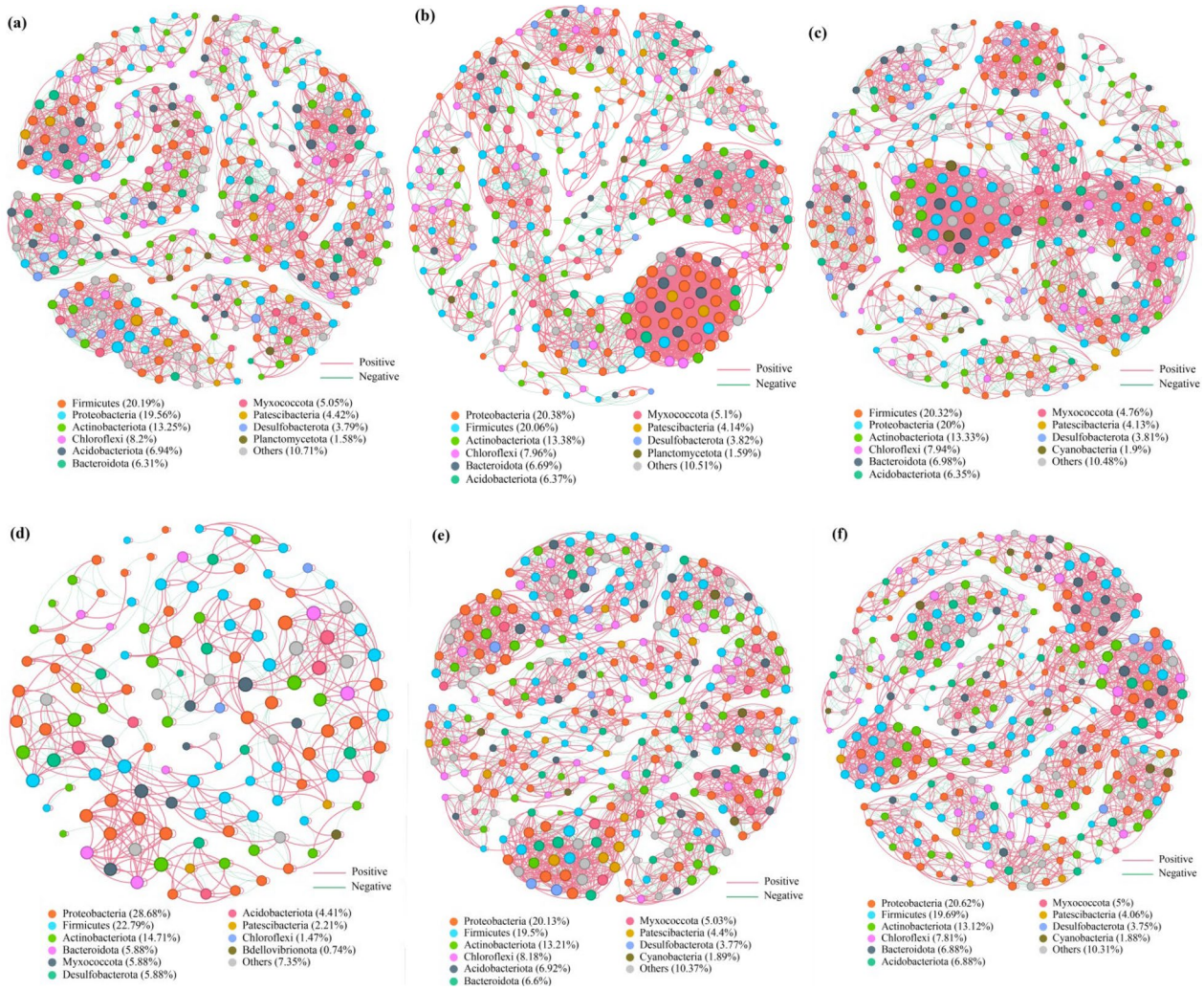


Fig. 6 The co-occurrence network of bacterial communities. **a** FCK; **b** FBC; **c** FFM; **d** ACK; **e** ABC; **f** AFM. Each node in the network represents a microbial genus, and different node colors represent different phyla. The lines connecting nodes (edges) represent positive (red) or negative (green) co-occurrence relationships

Actinobacteriota, Chloroflexi, Acidobacteriota, and Myxococcota, accounting for 79.50–88.23% of the total number of nodes. Proteobacteria and Firmicutes were the most dominant phyla in the network diagrams for all treatments and played a pivotal role in the overall network. Generally, complex microbial co-occurrence networks with elevated connectivity demonstrate greater resilience to environmental stresses than simpler, less-connected networks (Wang et al. 2021b). In this study, the flooded treatments significantly enhanced the number of nodes, edges, and average degree compared with the aerobic treatments. This indicated that the co-occurrence network under flooded conditions exhibited greater complexity and stronger associations. Flooding substantially restricts oxygen diffusion and intensifies soil redox gradients, shifting microbial energy acquisition from using oxygen as the terminal electron acceptor to greater reliance on stepwise redox processes and the transfer of metabolic intermediates. The resulting biogeochemical environment further stimulates metabolic complementarity and cooperative interactions among distinct functional groups, as manifested by increased edge density and mean degree centrality in microbial interaction networks (Anantharaman et al. 2016; Zhao et al. 2022). The clustering coefficient, which indicates the tendency of organisms to form clusters with high-density connections, was lower in the co-occurrence network under flooding conditions (0.768–0.804) than in aerobic conditions (0.789–0.806). All the treatments

exhibited modularity indices exceeding 0.4, thus confirming their modular network structures (Kong et al. 2019). Generally, higher modularity is associated with greater community stability because the loss of taxa within a module prevents its effects from spreading to the rest of the network (Fabbrini et al. 2023). Notably, FMBC addition resulted in the highest modularity under both water regimes, suggesting that FMBC integration fortifies the bacterial community network, conferring greater structural stability and enhanced resistance to environmental disturbances.

BC and FMBC amendments significantly enhanced the bacterial co-occurrence network complexity relative to the controls, as evidenced by increased nodes, edges, and average degree. These findings align with the reported biochar-induced shifts in bacterial network topology (Li et al. 2024). The strongest positive correlation was observed after FMBC addition (FFM and AFM treatments), likely because FMBC immobilized soil Cd and Hg, which may have alleviated competition and facilitated the mutualistic relationship among the bacterial communities (Yan et al. 2023). FMBC appears to be associated with shifts in microbial co-occurrence patterns, suggesting changes in ecological interactions. FMBC application increases SOC and resource availability, promoting mutualistic interactions among microorganisms (Wu et al. 2026).

In addition, the top 10 taxa ranked by degree were defined as keystone taxa (Table S4). Compared with the ACK and ABC treatments, the AFM treatment promoted Planctomycetota to emerge as a new keystone taxon within the microbial network. As a typical group of metal-resistant bacteria, Planctomycetota has been verified to effectively alleviate heavy metal toxicity by secreting extracellular polymeric substances (EPS), polysaccharides, and proteins (Li et al. 2021b). Compared with the FCK and FBC treatments, the FFM treatment enriched *Deinococcus*, *Herbaspirillum*, and *Novosphingobium*. Previous studies have shown that *Deinococcus* has been widely used in heavy metal pollution remediation because of its strong stress resistance, high tolerance to heavy metals, and its capacity for pollutant adsorption and degradation (Jin et al. 2019). *Herbaspirillum* has the ability to tolerate Cd toxicity and produce siderophores (Li et al. 2023), while *Novosphingobium* also exhibits strong tolerance to heavy metals (Ran et al. 2024). The enrichment of these keystone taxa indicates that FMBC addition not only reshaped the microbial co-occurrence network structure, but may also further improve the remediation capacity of soil for Cd and Hg contamination by enriching functional microbial groups with heavy metal tolerance and remediation potential.

3.5 Integrated pathways governing Cd and Hg accumulation in rice grains

To further clarify the effects of selected factors on the transformation and migration of Cd and Hg in the soil–rice system, PLSPM was used (Fig. 7). The model was applied to evaluate the effects of water management practices and FMBC application on Cd and THg accumulation in rice grains. The goodness of fit (GoF) of the models were 0.6788 (Grain–Cd) and 0.6854 (Grain–THg), respectively. The model indicated that the contents of DTPA–Cd (path coefficient = 0.4905), $Na_2S_2O_3$–Hg (path coefficient = 0.7413), and IMP (path coefficient = −0.5240 and 0.3832) had direct and substantial effects on grain Cd and THg concentrations. Water management and FMBC application can influence Cd and Hg availability in the soil and IMP formation on the root surface. In addition, they may exert indirect effects by altering soil properties (pH, Eh, CEC, SOC, and DOC) and bacterial communities, ultimately modulating Cd and Hg accumulation in rice grains. The total effect of water management on grain Cd and THg accumulation was substantially greater than that of FMBC. The water treatment had an indirect negative effect on grain Cd, whereas it had a significant positive effect on grain THg. These findings are consistent with those of previous studies. Liu et al. (2024) reported that flooding reduces the Cd content in brown rice primarily by regulating the availability of Cd, Fe, and Mn in the soil, the formation of iron plaques, and the Cd concentration in the xylem sap. Conversely, under flooded conditions, the anaerobic environment in paddy fields can enhance Hg mobilization into the soil solution and promote Hg methylation by anaerobic microbes, increasing the bioavailability of Hg and MeHg in rice plants (Wang et al. 2014; Li et al. 2022c).

A conceptual model illustrating the key pathways for the synergistic reduction of Cd and Hg accumulation by combined FMBC application and water management is presented in Fig. S3. The model elucidates the underlying mechanisms through the following interconnected pathways: (a) direct immobilization by FMBC: Cd and Hg are immobilized through the porous structure of FMBC, oxygen-containing functional groups, and Fe/Mn oxides; (b) water management-driven soil geochemistry: flooding creates a low-Eh environment that promotes Cd precipitation (CdS) but enhances Hg mobility and methylation potential; in contrast, aerobic conditions inhibit Hg methylation but increase Cd bioavailability; (c) root surface IMP formation: the combination of flooding and FMBC promotes the formation of IMP on root surfaces, which acts as a barrier to Cd uptake; (d)

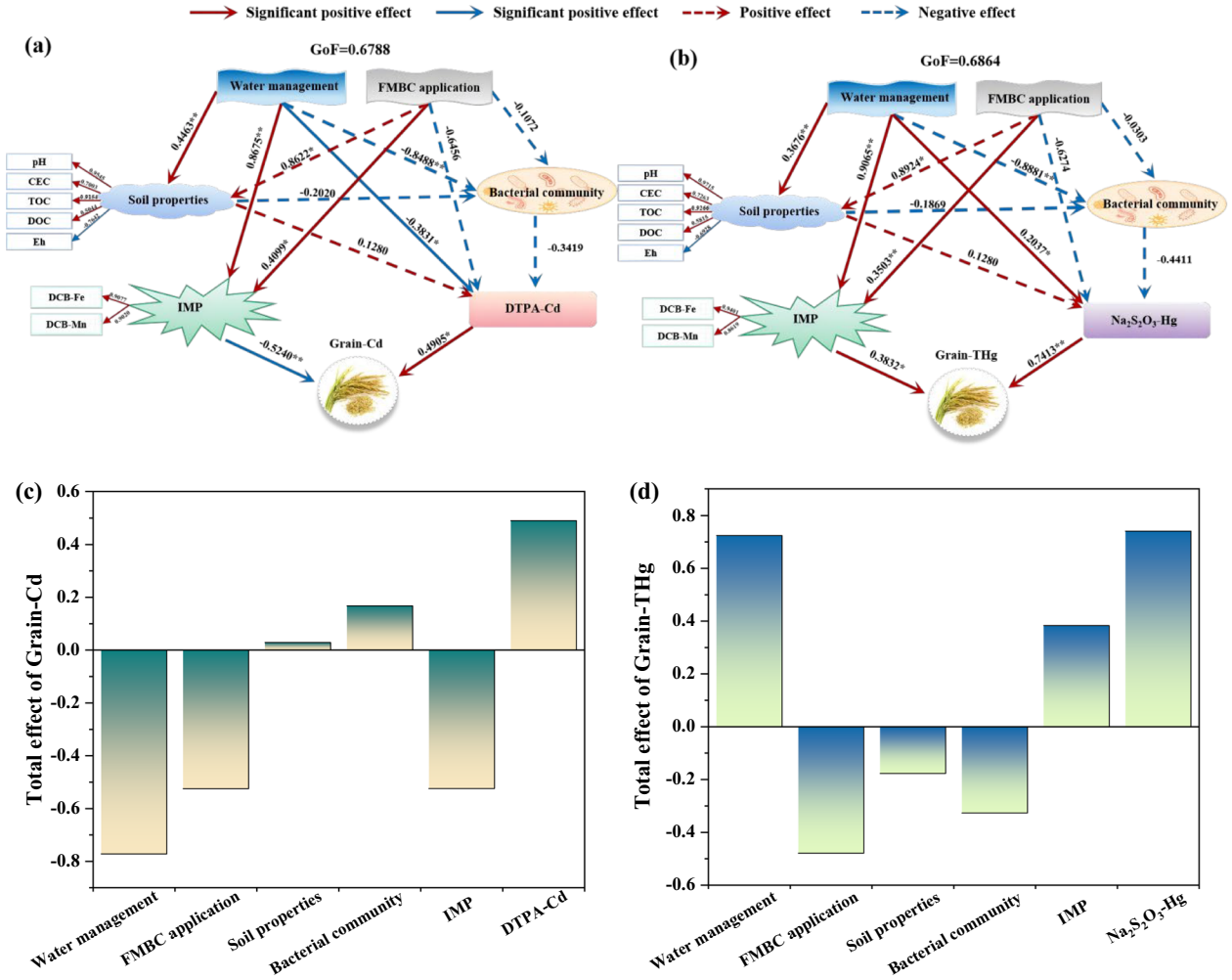


Fig. 7 The partial least squares path model (PLSPM) describing the effects of key factors on Cd (**a**) and Hg (**b**) concentrations in rice grain; The total standardized effects (sum of the direct and indirect effects) of each path to Grain-Cd (**c**) and Grain-THg (**d**). Soil properties refer to soil pH, CEC, TOC, DOC and Eh. IMP represents DCB-Fe and DCB-Mn concentrations. Bacterial community data are derived from the relative abundance of top 15 bacterial phyla. Red and blue represent negative and positive correlations, respectively. Numbers on arrows are standardized path coefficients, with solid lines indicating significant impacts and dotted lines indicating non-significant impacts. *, $P < 0.05$; **, $P < 0.01$. Model was assessed using Goodness-of-Fit (GOF), yielding GOF values of 0.6788 (Cd) and 0.6864 (Hg)

modulation of microbial-mediated processes: continuous flooding enriches *hgcA*-containing methylating microorganisms, such as *Geobacter* and *Desulfovibrio*, while FMBC reduces the methylation risk by decreasing the pool of bioavailable Hg.

4 Conclusions

The remediation of Cd and Hg co-contaminated paddy soils is fundamentally constrained by their opposing responses to redox states. Our results demonstrated that continuous flooding effectively limits Cd uptake by promoting IMP formation on the root surface, while increasing Hg mobility and stimulating *hgcA*-mediated Hg methylation under anaerobic conditions. Conversely, aerobic irrigation suppresses Hg methylation but enhances Cd bioavailability. The application of FMBC effectively functioned as a complementary amendment, mitigating metal-specific risks intensified by individual water regimes. Under flooded conditions, FMBC addition decreased the relative abundance of methylating microorganisms, and restricted grain Cd accumulation to a safe level of 0.05 mg kg^{-1}. Under aerobic conditions, the integration of FMBC restricted aerobic-induced Cd uptake and further immobilized inorganic Hg, synergistically

achieving the lowest grain THg (0.02 mg kg^{-1}) and MeHg (6.89 μg kg^{-1}) concentrations. In summary, our findings provide a practical framework for tailoring water-amendment combinations to mitigate the risks posed by Hg and Cd in polluted paddy soils.

However, this study was conducted under controlled pot conditions, which may not fully capture field-scale dynamics. Field applicability faces practical challenges, including the economic costs of FMBC scale-up, meteorological disruptions to strict water regimes, and the long-term stability of the amendments under natural weathering. Future research should focus on multi-year field trials to verify the optimal combination of water management regimes and FMBC application, aiming to simultaneously and effectively inhibit the bioaccumulation of Hg and Cd in grains.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s42773-026-00616-w.

Supplementary material 1.

Author contributions

Tong Sun: Conceptualization, Investigation, Writing–original draft preparation. Wenhao Yang: Visualization, Validation. Yuebing Sun: Supervision, Writing–reviewing and editing, Funding acquisition. Lin Wang: Visualization, Methodology. Xuefeng Liang: Conceptualization, English-editing. All authors read and approved the final manuscript.

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Data availability

The data that support the findings of this study are available from the corresponding author upon request.

Declarations

Competing interests

We have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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